[METRIC-SI]
AA-O-00230D/GEN (GSA-FSS)
July 1, 1993
INTERIM REVISION
FED. SPEC. AA-O-230C/GEN
July 21, 1989

INTERIM FEDERAL SPECIFICATION

OFFICE FURNITURE: EXECUTIVE TYPE, WOOD, TRADITIONAL (GENERAL SPECIFICATION)

This Proposed Federal Specification was developed by the Furniture Commodity Center, Federal Supply Service, General Services Administration, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification AA-O-230A dated October 7, 1964.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the general requirements for executive type, wood, traditional office furniture. Specific requirements, figures and drawings, and other information is in the detailed specification AA-O-00230/3 Costumer, Executive Office, Wood Traditional.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Military Standards.

MIL-STD-105 - Sampling Procedures and Tables for Inspection for Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

FSC 7110

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards.

ASTM D 905 - Standard Test Method for Stength Properties of Adhesive in Shear by Compression Loading.
ASTM D 1211 - Method of Test for Temperature Change Resistance of Clear Nitro-cellolose Lacquer Films Applied to Wood.

(Copies may be obtained from the American Society for Testing and Materials 1916 Race Street, Philadelphia, PA 19103).

3.0 REQUIREMENTS

3.1 Regulatory Requirements. The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

3.2 Materials. The materials used in construction of the furniture shall be as follows:

3.2.2 Adhesives. The adhesives to be used and their permitted application shall be as shown in table I.

TABLE I. Adhesives

Application	Kind of adhesives
All glue blocks, assembly joints, glued, laminated stock.	Any adhesive that will develop an average block shear strength of not less than 19 300 kPa when tested as specified in 4.5.2.

3.2.2 Wood.

3.2.2.1 Wood species. Wood permitted are those listed in table II.

TABLE II. Wood species

Exposed Solid Parts 1/
Cherry
Mahogany 2/
Walnut
Andiroba (Carapa guianensis)

1/ Only one specie may be used in any one unit

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2/ Khaya/African or Swietenia/American.

3.2.2.2 Solid wood requirements. Wood used for exposed parts, shall be bright, well sanded, and free from brashness, discolorations, worm holes, splits, shake, or other defects which affect the appearance or which may affect the serviceability of the finished product.

3.2.2.3 Wood seansoning. All wood, shall be uniformly kiln-dried to a 5 to 8 percent moisture content as determined by the oven-drying method. After drying, it shall be held in temperature and humidity controlled storage for a sufficient time to ensure uniform moisture distribution among the individual pieces.

3.2.3 Hardware.

3.2.3.1 Face hardware All face hardware such as coat hooks, shall be made of cast brass. The castings shall be true without air or sand holes or other defects. All ornamentation on face of the hardware, such as mouldings, beadings, or turnings, shall have a clean and even surface. All surfaces shall be smoothly and evenly finished. Hardware shall receive minimum one coat of baked synthetic clear finish.

3.2.3.2 Nail glides. Glides shall be 22 mm minimum diameter, rubber cushioned nail type. The base shall be brown high impact plastic.

3.2.3.3 Other hardware. All screws, whose heads are exposed on brass hardware shall be brass.

3.3 Construction.

3.3.1 Design and construction. The design and construction of the items shall be as specified herein and in the applicable detailed specification. In any case where there is a conflict between the text of the specification and the figures, illustrations, or drawings, the text shall prevail.

3.3.2 Tolerances. Where tolerances are not specified, the following tolerances will be permitted:

Overall dimensions: plus, minus 6 mm.
Solid wood part dimensions: plus 4 mm, minus 2 mm.

3 3 3 Direction of grain The grain direction shall be as shown in the applicable figures

3 3 4 Joinery All joints shall be true, securely glued (3.3.1), and unless otherwise specified, shall be tongue and grooved or mortised and tenoned Starved or insufficiently glued joints will not be acceptable

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3.3.5 Dowels. Dowels shall be made of maple, birch, hickory, or beech, dried to a moisture content of 3 to 5 percent. They shall have longitudinal or spiral grooves and shall be beveled on each end.

3.3.6 Built-up stock. Built-up stock used for exposed components in the construction of the furniture shall consist of not more than three full length pieces, each of which shall be approximately the same thickness They shall be carefully matched for color and figure.

3.4 Finishing.

3.4.1 Finish, exposed wood parts and surfaces. All surfaces exposed to view in normal use shall be finished, sanded smooth with all corners and edges eased. Exposed wood finish shall match walnut standard sample FSS-L-01002 (6.3) for color and appearance. The natural grain of the wood shall not be clouded by the finishing materials. Bleaching agents or materials shall not be used. The application of materials, drying time, sanding, cleaning, and rubbing shall be controlled to produce items of uniform finish without sags, runs, orange peel or other defects detrimental to a smooth quality appearance. Final finish shall be satin. Finish shall meet test requirements in 4.5.1.

3.4.2 Recommended procedure for finishing exposed wood surfaces.

1. Sap satin to equalize color.
2. Apply selected satin to produce color specified.
3. Wash coat with adhesive sealer.
4. Scuff sand, using sandpaper grit of fineness needed to produce proper surface.
5. Fill to degree necessary to match standard sample.
6. Seal with adhesive type sealer.
7. Sand, using sandpaper grit of fineness needed to produce proper surface
8. Shade to tie in color variation, if necessary.
9. Top coat with lacquer (two coats).
10. Rub with material necessary for final finish to conform to standard sample.

NOTE: Operation number 8 may be performed where shown or after operation number 5.

3.4.3 Finish, unexposed parts and surfaces. All unexposed exterior parts shall be machined smooth and cleaned. These parts and surfaces shall receive one coat of lacquer sealer and one coat of lacquer (3.1.3)

3.5 Identification marking. Each unit shall be permanently and clearly marked with contrasting ink in an inconspicuous place with the Federal Stock Number, contract number, this specification number, the year and month of manufacture, and the manufacturer's name or trademark.

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3.6 Workmanship. Method of machining, construction, joinery, gluing, and assembly, shall conform to best commercial practice for furniture of this style. All surfaces exposed to view in normal use shall be smoothly machined and sanded. Units shall be free of splinters, sharp corners, and sharp edges to prevent injury to personnel or damage to their clothing. The natural grain of the wood shall not be clouded by the finishing materials. The application of materials, drying time, sanding, and cleaning shall be controlled to produce items of uniform finish without sags, runs, orange peel, overspray or other defects detrimental to a smooth quality appearanace.

4. QUALITY ASSURANCE PROVISIONS

4.1 First article inspection and testing. The required samples shall be inspected and tested by the contractor for all the requirements of the contract. A record of this inspection and test, including certificates of conformance for materials, shall be submitted to the government for approval. The Government reserves the right to witness the contractor's inspection and test.

4.1.1 First article approval. The requirements and samples required for first article inspection and testing shall be as specified in the contract. Each required item shall be examined and tested as specified herein for all the requirements of this specification. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. A new first article sample is required for each new contract. Previous first article samples are not acceptable. Approval of first article sample(s) does not relieve the contractor of the responsibility of complying with all applicable specification provisions. Manufacturer shall maintain first article sample as a manufacturing standard unitl the last order is shipped, received and accepted by the ordering activity.

4.2 Lot acceptance tests. Lot acceptance tests shall be conducted under the supervision of the Government Inspector at the manufacturer's plant. In the event the manufacturer does not have the facilities to perform the required tests, he shall be submitted to the Government Inspector upon receipt from the laboratory. All examination and testing shall be in accordance with Section 4 of the specification.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated.

4.4 Inspection levels and acceptance quality levels (AQL's) Inspection levels and acceptable quality levels expressed in defects per hundred units shall be as shown in table IV.

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TABLE IV.Acceptable quality levels in accordance with MIL-STD-105

For examination in	Inspection Level	AQL's Major	AQL's Total
4.4.1	II	4.0	25.0
4.4.2	II	4.0	
4.4.3	S-2	4.0	
4.4.4	S-2	4.0	

4.4.1 Visual examination. Examination of the end item(s) shall be in accordance with the classification of defects specified in table V and acceptable quality levels (AQL's) specified in 4.3 herein The lot size shall be all units of the same type offered for inspection at one time. The sample unit for this examination shall be one complete item.

TABLE V. Classification of end item defects

Examine	Defect	Classification Major	Classification Minor
Finish and appearance of exposed wood parts visible in normal use position.	Does not match Standard reference panel with respect to color and appearance.	X	
	Discoloration (bleeding) over 25 square mm.		X
	Spotted, affecting appearance.		X
	Light or dark colored areas, affecting appearance.	X	
	Area of no lacquer or varnish film over 25 square mm in area.	X	
	Glue stain or glue line clearly showing affecting apperance.		X
	Excess glue not removed.		X
	Finish peeling, flaking, blistered, cracked, checked or wrinkled	X	
	Finish cut or rubbed through to wood.		X

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TABLE V. Classification of end item defects (cont)

Examine	Defect	Classification Major	Minor
	Open grain. Finishing material not filling pores.		X
	Orange peel, affecting appearance.		X
	Damage or scratch, over 25 mm sq. or 25 mm L	X	
	Finish tacky, i.e., not completely dry		X
	Rough or gritty surface or foreign matter imbedded in finish.		X
	Light colored edge not touched up.		X
	Burn-in or repairs not neatly accomplished.		X
	Unstained (white) wood visible when item is in normal use postion.		X
Finish of exterior wood parts not visible in normal use.	Finish not applied where required.		X
	Stain or sealer coat omitted where required.		X
Finish of hardware.	Does not match standard finish.	X	
	Component tarnished or stained, affecting appearance.		X
	Rusted or corroded.		X
	Not coated with protective finish.		X
	Species of wood not as specified	X	
	Component not fabricated of one piece solid wood where required.	X	

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TABLE V. Classification of end item defects (cont)

Examine	Defect	Classification Major	Minor
Quality of solid etc. (cont.)	More than one species of wood used for exposed wood components of any one item.		X
	Direction of grain not as specified.		X
	Check or split over 25 mm in length.	X	
	Wood warped, affecting appearance or serviceability.		X
	Loosened grain or shake on exposed surface.		X
	Loosened grain or shake on unexposed surface.		X
	Honeycomb.		X
	Loose or unsound knot.		X
	Hole on exterior surface affecting appearance.	X	
	Glue line in built up stock open over 25 mm in length.		X
	Glued up pieces uneven in built-up stock.		X
	Wave of uneven sanding.		X
	Machine mark on exposed surface affecting apperance.		X
	Any mineral streak on exposed surface over 0.5 x 100 mm or 1 x 50 mm		X
	Any manufacturing defect or defect native to the wood filled in with plastic wood or other filler material.	X	

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TABLE V. Classification of end item defects (cont)

Examine	Defect	Classification Major	Minor
Quality of metal components (except attaching implements)	Not fabricated of material specified.	X	
	Broken, cracked or fractured.		X
	Dented or damaged.		X
	Nicked, pitted, malformed or bent, affecting appearance.		X
	Machine or die marked, affecting appearance.		X
	Contains hole where not required, does not have one where required.		X
	Fin, burr, or sharp edge which may cause personal injury or damage to clothes.		X
Workmanship and attachment of screws, bolts, nuts, washers, and rivets.	Any attachment device not type or size as specified.		X
	Material not as specified.		X
	More than one screw missing.		X
	Spacing of screws not as specified.		X
	Screw point visibly bulging wood.		X
	Wood at insertion point not counter sunk when specified.		X
	More than one screw, bolt, nut washer or rivet missing.		X
	More than one bolt, screw, or nut not tightened		X
	Threads on nuts or bolts stripped.		X

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TABLE V. Classification of end item defects (cont)

Examine	Defect	Classification Major	Minor
Workmanship and attachment of screws, bolts, nuts, washers, and rivets (cont.)	Screws, bolts, or rivets broken off.		X
	Screw or bolt in wrong location.		X
Construction, assembly, and attachment of unexposed and exposed parts.	Not as specified or not as shown on drawings or figures.	X	
	Component not size or shape specified.		X
	Component not type or design specified.		X
	Component not assembled as specified.		X
	Component missing.	X	
	Component in wrong location, except attaching implements.		X
	Component loose or insecurely fastened where rigidity is required.		X
	Component out of alignment or out of square affecting serviceability.		X
	Poor fit of component part or on matting part.		X
	Wood component not glued as specified.		X
	Solid wood used in place of veneer or vice versa		X
	Wood component damaged or not properly machined.		X
	Molding, fluting routing, or beading not uniform or uneven.		X

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TABLE V. Classification of end item defects (cont)

Examine	Defect	Classification Major	Classification Minor
Construction, etc. (cont.)	Component not chamfered where required.		X
	Edges not "eased off" where required.		X
	Any loose or unsound glued joint.		X
	Wood joint not as specified.		X
	Glue joint open over 1 mm on exposed part		X
	Glue joint open over 3 mm on unexposed part.		X
	Legs or base uneven.		X
Identification marking.	Missing, incorrect or illegible.		X
	Text incomplete.		X
	Marking in wrong location.		X
	Size of lettering not as specified.		X
	Color of marking not black.		X

4.4.2 Assembly examination. Examination of assembly shall be in accordance with paragraph 4.2 in the detailed specification.

4.4.3 Examination of dimensions. Any dimensions that is not as specified in figures or text shall be classified as a defect.

4.4.4 Examination of preparation for delivery. An examination shall be made to determine compliance with requirements of section 5. The sample unit shall be one shipping container, identical to each unit being shipped, fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105 (see 4.3).

4.5 Tests.

4.5.1 Finish tests and requirements. (See 3.4.1) The following tests (except UV test) shall be performed on a sample panel finished in the same manner as units are finished in production

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Perform all finish tests at first article inspection and again one year after the start of the schedule contract period. Perform tests (a, b, and g) once a month during the contract period. All test panels shall be produced from finish materials currently being used in production. All samples tested shall meet the following test requirements. Actual test results are required. Certifications are not acceptable.

(a) Cold check. Age panel one week. Follow ASTM D 1211 test procedure. After exposure of 10 cycles, there shall be no checking or cracking. True film checking is one or more wavy or straight lines which cross the grain and do not appear related to grain structure. When a panel displays veneer checks, check running parallel to the grain, or glue line fracture, the panel shall be disregarded and another panel tested.

(b) Cold print. Age panel 48 hours. A 50 x 50 mm (approx.) piece of duck material (canvas) 280 g/m^2, minimum count of 23.6 x 22.8 yarns per cm, shall be placed on the panel. Place a 455 g weight on the material for 24 hours at 24 degrees C. Bottom (contact surface) of the weight shall be flat and have a 28 mm diameter. After the weight is removed, light polishing with a soft cloth and liquid polish must remove any imprint.

(c) Ultra violet light resistance. Spray one-half of a 150 mm x 230 mm white porcelain panel with two coats of the lacquer to be tested. Allow panel to air dry for at least 18 hours. Place panel 305 mm from an ultraviolet light (Uviarc 420-U1 light with UA-3 lamp) with a 3.18 mm thick pane of window glass placed between the lamp and the panel, and expose for 24 hours. After exposure, spray the other half of the panel in the same manner and allow to dry for at least 4 hours. Compare the two finishes. There shall be no more than a slight darkening of the UV light exposed film when checked under cool white fluorescent light.

(d) Toughness and adhesion. Perform test for toughness and adhesion on a sample panel finished in the same manner as the furniture using Organic Coating Adhesion Tester, Model No, 1001 in accordance with manufacturer's instructions. Mar the panel both parallel and perpendicular to the grain. Film must conform to resulting indentation. Whitening (film separation) or cracking is not acceptable.

Organic Coating Adhesion Tester Model No. 1001 is available from U.S. Testing Company, Inc. Instrument Marking Division, 1415 Park Avenue, Hoboken, NJ 07030.

4 5 2 Test for adhesives Test for adhesives shall be as shown in table VII

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TABLE VII. Test for adhesives

Component	Characteristic	Requirement reference	Test method
Adhesive	Block shear test	3.1.2	ASTM D 905

Rerun test if the average shear strength of all samples is below 19 300 kPa; there is a 10 percent or greater difference between high and low specimen values, and at least one test specimen broke at more than 19 300 kPa

Disregard a test specimen in computing the average if it breaks at less than 19 300 kPa; and it has 50 percent or more wood failure.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing, Marking. Package, pack, and mark shipping containers in accordance with the contract or order.

6. NOTES

6.1 Intended use. The traditional executive furniture items are designed for use in executive offices, courtrooms, libraries, and conference rooms of Federal agencies.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Title, number, and data of the detailed specification.

6.3 Standard plastic laminate sample. A standard plastic laminate sample (walnut FSS-L-01002) may be obtained from:

National Furniture Center
Engineering Division
GSA-FSS-3FNEW
Washington, DC 20406

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6.4 SI - English unit equivalents.

1 m^2 (square meter)	= 1.19617 $yard^2$
1 kg/m^3 (kilogram/cubic meter)	= 0.06242 lb.(mass)/ft^3
1 mm (millimeter)	= 0.03937 inch (thickness of one dime)
1 m(meter) = 1 000 mm	= 1.0936 yard (39.7 in)
1 N(Newton)	= 0.225 lb. (force)
1 kg(kilogram)	= 2.2 lb.(mass)
1 g(gram)	= 0.0022 lb. (mass)
1 g(gram)	= 0.03527 oz.(mass avoirdupois)
1 kPa(kilo Pascal)	= 0.14514 lb(force)/in^2(PSI)
(C^o x 9/5) + 32 (Celsius)	= F^o (Fahrenheit)
1 g/m^2 (gram per square meter)	= 0.02949 oz/yd^2 or 0.04426 oz/linear yard (54" W basis)

To convert SI units to English units, multiply SI measurement by the appropriate English conversion factor listed above. See example on next below:

900 mm X 0.03937 in./mm = 35.43 inches

To convert Celsius temperature to Fahrenheit temperature use the above conversion equation. See example below:

(20°C X 1.8) + 32 = 68°F

Preparing Activity
GSA/FSS

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